Finite CMB compensation budgets in a bounded nine-moment foreground class

Anonymous methods and results note

Seven retained passband responses give a full-row-rank matrix for nine signed, second-order synchrotron and dust moments in each of six fixed response conventions. In the unrestricted linear class, the CMB response can consequently be represented by foreground moments in every one of the 24 signed $E_2/E_3/B_2/B_3$ coordinates. A declared joint quadratic bound changes that statement quantitatively: the least moment budget for a unit CMB response is 3.117720–3.118388 target-amplitude units in one common column-normalized metric. A single foreground can compensate a target tensor displacement of norm at most ρ/ℓ_M, while two bounded foregrounds can make two targets separated by as much as $2\rho/\ell_M$ agree on the same synthetic data. For arbitrary specified frequency data, the complete 24-dimensional compatible target set is a calculable ball or is empty. The raw-coordinate identity metric changes the unit cost to 59.572829–59.592192, displaying explicit metric sensitivity. These are finite algebraic results for the declared free signed-moment class. No sky tensor, physical moment bound, joint noise law, or native solver output is used.

I. PHYSICAL MECHANISM AND FROZEN FINITE RESPONSE

A passband integrates spectral components before their angular coefficients are estimated. Spectral changes in a component's index or temperature therefore add frequency response columns to its leading spectrum. If enough independently variable moment maps are admitted, their channel responses can reproduce the CMB's calibrated response. A bound on their joint amplitude restores a finite compensation cost, even when the unrestricted response matrix spans all channels. That cost depends on which moment amplitudes the bound regards as comparable.

This note consumes the already computed, released-bandpass finite matrices in the retained R31 result identified in the accompanying claim-to-source map; it performs no passband integration or map analysis. The seven frequency rows, in order, are 030, 044, 070, 100, 143, 217, and 353PSB. The nine columns are

$$(s, s_\beta, d, d_\beta, d_t, s_{\beta\beta}, d_{\beta\beta}, d_{\beta t}, d_{tt}). \tag{1}$$

Here $t = (T - 20\,\mathrm{K})/(20\,\mathrm{K})$; the retained pivots are $\beta_s = -3$ in Rayleigh–Jeans temperature, $\beta_d = 1.5$, $\nu_s = 30\,\mathrm{GHz}$, $\nu_d = 353\,\mathrm{GHz}$, and $T_{\mathrm{CMB}} = 2.7255\,\mathrm{K}$. Taylor factorials are absorbed into the signed moment coefficients. In particular, the second temperature response uses $T_d^2\partial_T^2$, with $T_d = 20\,\mathrm{K}$. The two LFI frequency conventions A/B and three HFI transmission thresh-

olds $10^{-7}, 10^{-8}, 10^{-6}$ are retained as six separate finite response models. Their envelope is not a calibrated passband uncertainty.

The full target carrier has $5 + 7 + 5 + 7 = 24$ real coordinates: E_2, E_3, B_2, B_3, with $m = 0$ followed for each positive m by $(\sqrt{2}\,\mathrm{Re}, -\sqrt{2}\,\mathrm{Im})$. Write the frequency-major data $Y \in \mathbb{R}^{7\times 24}$, target $s \in \mathbb{R}^{24}$ and moment array $U \in \mathbb{R}^{9\times 24}$ as

$$Y = as^{\mathsf{T}} + FU, \qquad a = (1, \ldots, 1)^{\mathsf{T}} \in \mathbb{R}^7, \qquad \mathrm{vec}_{\mathrm{row}}(Y) = (a \otimes I_{24})s + (F \otimes I_{24})\,\mathrm{vec}_{\mathrm{row}}(U). \quad (2)$$

All coordinates, including derivative moments, are free and signed. s, U, and Y use the same target harmonic-amplitude unit, taken here as $\mu\mathrm{K}_{\mathrm{CMB}}$ when a unit is needed. F and the metric scales below are dimensionless under the retained response normalization. There is no observed Y in this experiment.

The primary class uses a single metric across all six branches:

$$\begin{aligned} d = (&1.261652382369369,\ 0.1777139521270542,\ 1.1309894216176297,\\ &0.09451923139372441,\ 0.04109240642871376,\ 0.1151042544103929,\\ &0.07145402059880916,\ 0.022520789584731377,\ 0.09386151314750793),\\ M = \mathrm{diag}(&d_1^2, \ldots, d_9^2), \qquad \|U\|_M^2 := \mathrm{tr}(U^{\mathsf{T}} M U) \le \rho^2. \end{aligned} \quad (3)$$

The d_i are the stored column norms of the nominal A, 10^{-7} matrix. Thus ρ is a jointly applied computational moment budget in the target-amplitude unit, not a measured foreground prior. We also compute $M = I_9$ as an explicitly different metric. Neither metric imposes physical positivity or cross-moment coupling.

II. MINIMUM COMPENSATOR AND SINGULAR DIRECTIONS

For a full-row-rank F and any symmetric positive-definite M, define

$$K = FM^{-1}F^{\mathsf{T}}, \qquad c_M = M^{-1}F^{\mathsf{T}}K^{-1}a, \qquad \ell_M^2 = a^{\mathsf{T}}K^{-1}a. \quad (4)$$

The 7×7 matrix K is positive definite. Lagrange multipliers for $\min_{Fc=a} c^{\mathsf{T}}Mc$ give $Mc = F^{\mathsf{T}}\lambda$, $K\lambda = a$, hence Eq. (4). If $Fc = a$, then $c = c_M + n$ with $Fn = 0$, and

$$c^{\mathsf{T}}Mc = \ell_M^2 + n^{\mathsf{T}}Mn. \quad (5)$$

This proves the unique minimum and retains the two-dimensional coefficient nullspace. It also proves that the signed compensation $\Delta U = -c_M v^{\mathsf{T}}$ has minimum budget $\ell_M \|v\|_2$ for any complete target displacement $v \in \mathbb{R}^{24}$.

TABLE I. Minimum cost ℓ for a unit target tensor displacement and the corresponding one-foreground radius per unit budget. M is the fixed common metric of Eq. (3); I is a separately declared raw-coordinate identity metric. A pairwise separation has twice the displayed radius. The branch labels retain the LFI convention and HFI threshold.

Branch	ℓ_M	$1/\ell_M$	ℓ_I	$1/\ell_I$
A_BT_1e-07	3.118381	0.320679	59.592105	0.016781
A_BT_1e-08	3.118381	0.320679	59.592092	0.016781
A_BT_1e-06	3.118388	0.320678	59.592192	0.016781
B_BT_1e-07	3.117720	0.320747	59.572842	0.016786
B_BT_1e-08	3.117720	0.320747	59.572829	0.016786
B_BT_1e-06	3.117727	0.320747	59.572929	0.016786

To display which response directions incur the cost, factor

$$FM^{-1/2} = U_f \,\mathrm{diag}(\sigma_1, \ldots, \sigma_7) V_f^{\mathsf{T}}, \qquad \ell_M^2 = \sum_{i=1}^{7} \frac{(u_i^{\mathsf{T}} a)^2}{\sigma_i^2}. \tag{6}$$

The first seven rows of V_f^{T} are orthonormal directions in whitened coefficient coordinates; $M^{-1/2} v_i$ are their original-coordinate versions. The complete singular values, projections, squared budget contributions, and vectors for each branch and both metrics are generated in the machine-readable result. The minimum singular value in the common metric is about 0.00553–0.00554. For the nominal A branch, the seventh squared-budget contribution is 7.4391 of $\ell_M^2 = 9.7243$; the singular value alone would not show this CMB alignment. The smallest-to-largest singular-value ratio exceeds the retained rank-resolution cutoff 10^{-10} in all six branches.

Table I gives $\ell_M = 3.117720$–3.118388 across the six branches. In the raw metric, $\ell_I = 59.572829$–59.592192, agreeing with the previously stored raw-coordinate witnesses to relative 10^{-8}. The metric change is substantive: it defines a different feasible coefficient class. The small A/B and threshold variation within each metric is a finite-model sensitivity, without an inference about real response uncertainty.

III. COMPLETE COMPATIBLE TARGET SET AND FACTOR OF TWO

For specified Y, minimize the budget over U at each proposed s. Applying Eq. (4) separately to the 24 columns gives

$$U_s = M^{-1} F^{\mathsf{T}} K^{-1} (Y - a s^{\mathsf{T}}), \qquad J(s) = \mathrm{tr}\left[(Y - a s^{\mathsf{T}})^{\mathsf{T}} K^{-1} (Y - a s^{\mathsf{T}}) \right]. \tag{7}$$

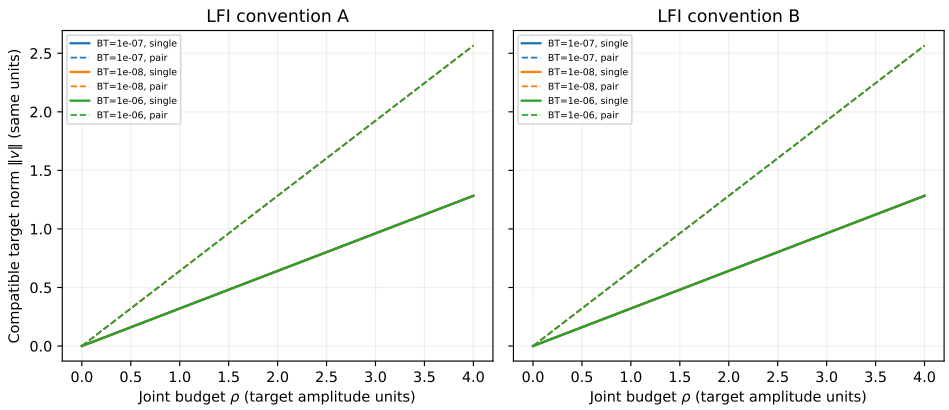


FIG. 1. Attainable full-tensor displacement norm as a function of the declared joint moment budget ρ under the same common metric in all six branches. Solid curves give the one-foreground limit ρ/ℓ_M; dashed curves give the pairwise maximum $2\rho/\ell_M$. Colors distinguish the three HFI thresholds in each LFI panel. Nearly overlapping curves reflect the computed response-convention sensitivity. These are algebraic feasible-set frontiers for a free signed-moment class, not observed CMB amplitudes or confidence limits. Figure source: `consumer.py`, frozen R31 matrices.

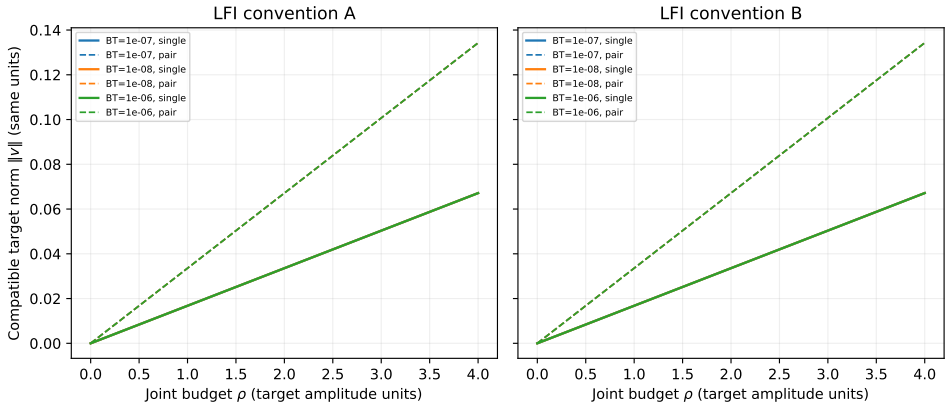


FIG. 2. The same frontiers after explicitly replacing the common column-normalized metric with raw-coordinate $M = I_9$. Axis units and the factor-of-two distinction are unchanged; the shift in slope exposes the dependence on the bounded class. This panel is a metric-sensitivity calculation, not a second physical prior.

Define the full signed center and its irreducible cost by

$$s_0 = \frac{Y^{\mathsf{T}} K^{-1} a}{\ell_M^2}, \qquad J_0 = \operatorname{tr}\left[(Y - a s_0^{\mathsf{T}})^{\mathsf{T}} K^{-1} (Y - a s_0^{\mathsf{T}})\right] \geq 0. \tag{8}$$

Expanding $J(s)$ about s_0 makes the cross term vanish by the definition of s_0, so

$$\boxed{\mathcal{S}_\rho(Y) = \{s \in \mathbb{R}^{24} : J_0 + \ell_M^2 \|s - s_0\|_2^2 \leq \rho^2\}.} \tag{9}$$

It is empty for $\rho^2 < J_0$, a singleton at equality, and otherwise a closed 24-dimensional ball centered at s_0 with radius $\sqrt{\rho^2 - J_0}/\ell_M$. Its shape matrix in the given coordinate frame is $\ell_M^2 I_{24}$. We evaluate J_0 from the residual in Eq. (8), avoiding subtraction of nearly equal quadratic forms. Equation (9) is a deterministic compatible set, not a posterior or confidence region.

The zero-data illustration $Y = 0$ has $s_0 = 0$ and $J_0 = 0$. One foreground can hide v exactly when $\ell_M \|v\| \le \rho$, with $U = -c_M v^{\mathsf{T}}$. For two foregrounds, each separately bounded by ρ, a common zero data table is obtained from

$$s_1 = v/2, \quad s_2 = -v/2, \qquad U_1 = -c_M v^{\mathsf{T}}/2, \quad U_2 = c_M v^{\mathsf{T}}/2. \tag{10}$$

Each budget equals $\ell_M \|v\|/2$. Conversely $F(U_2 - U_1) = a(s_1 - s_2)$ and the triangle inequality give $\ell_M \|s_1 - s_2\| \le \|U_2\|_M + \|U_1\|_M \le 2\rho$. Thus the factor of two is both necessary and attained. At fixed generic Y, the diameter is $2\sqrt{\rho^2 - J_0}/\ell_M$, when the set is nonempty. As $\rho \to \infty$ this finite obstruction disappears, consistently with the zero target quotient rank of the unrestricted nine-moment class. At $\rho = 0$, $U = 0$ and compatible data must be exactly $Y = as^{\mathsf{T}}$.

IV. NUMERICAL EXECUTION AND SCOPE

The consumer checks the exact SHA-256 of the retained R31 result before reading any matrix. For each branch and metric, it solves K by Cholesky, compares the minimum budget with an independently factored scaled SVD, checks $Fc_M = a$, the KKT equation, nullspace orthogonality and a positive norm increment. It exports each full 168×24 target operator A, 168×216 moment operator B, 216×24 compensator operator, 24×168 center map and 24×24 shape matrix. The synthetic controls act on all 24 basis vectors and a mixed signed vector, test boundary witnesses and 1% beyond-boundary rejection, and construct a generic Y with $J_0 > 0$ to test empty, singleton and ball sectors. Transporting the metric under nonzero column rescaling leaves K and the set invariant; a fixed orthogonal carrier rotation preserves the budget and set. All controls are generated in the same execution as the reported table, with numerical residuals in Table II.

The synthetic nonzero-data control has $J_0 \simeq 1.7174$ in the nominal branch. It realizes an empty set below $\sqrt{J_0}$, a singleton at equality, and radius 0.5 at $\rho = \sqrt{J_0 + \ell_M^2 (0.5)^2}$. This is a numerical witness for Eq. (9), not a sky realization. The smallest common-metric singular value and the complete basis residuals are shown in Table II; the exported binary64 coefficients, not only an internal intermediate, meet the residual gate.

The moment class permits signed independent harmonic and derivative coordinates. Physical

TABLE II. Smallest resolved singular value, relative $Fc = a$ residuals under both declared metrics, and largest relative full-tensor basis-action residual $\max_j \epsilon_j$ for the common-metric compensator. The residual gate is 10^{-10} for $Fc = a$ and 10^{-9} for the complete tensor actions.

Branch	$\sigma_{\min}(FM^{-1/2})$	$\|Fc_M - a\|/\|a\|$	$\|Fc_I - a\|/\|a\|$	$\max_j \epsilon_j$
A_BT_1e-07	0.0055441	2.26e-14	1.13e-12	2.26e-14
A_BT_1e-08	0.0055441	4.71e-14	7.16e-13	4.71e-14
A_BT_1e-06	0.0055440	5.11e-14	1.22e-12	5.11e-14
B_BT_1e-07	0.0055272	3.43e-14	2.17e-12	3.43e-14
B_BT_1e-08	0.0055272	4.19e-14	1.61e-12	4.19e-14
B_BT_1e-06	0.0055271	3.52e-14	1.03e-12	3.52e-14

component intensities, if positive, do not imply positivity of these harmonic or derivative coefficients. A physical version would need externally justified magnitude, coupling and truncation-remainder constraints, component-specific spatial responses, a real joint noise/calibration model, and a justified radius or prior. Those inputs are absent here. A nuisance with exactly the blackbody response remains confounded under this spectral model. We therefore make no statement about an observed quotient tensor, actual CMB amplitude, physical foreground filling fraction, calibrated exceedance, global tilt or Bianchi family. The retained status fields are $x = \text{NOT_MEASURED}$, $Q = \text{NOT_EVALUATED}$, $F_{\text{physical}} = \text{NOT_EVALUABLE}$, $\Pi = \text{NOT_CALIBRATED}$, and $G_F = \text{NOT_APPLICABLE}$ because frequency is not depth. Complete source identities and claim ceilings are given in the accompanying `CLAIM_SOURCE_MAP.md`.